AI.109

Embedded Systems

COURSE OVERVIEW

Computing on constrained hardware: microcontrollers, I/O, interrupts, and real-time considerations for robotics.

1

SESSION

60 MINUTES

ORIENTATION: COMPUTERS IN THINGS

KEY TOPICS

- Embedded vs. general-purpose systems
- Microcontroller architecture
- The embedded software stack
- Role in robotics

TEACHING FOCUS

Understand computing inside physical devices.

KEY TAKEAWAY Embedded systems are computers that interact with th

EMBEDDED SYSTEMS

2

SESSION

60 MINUTES

MICROCONTROLLER INTERNALS

KEY TOPICS

- CPU, memory, and buses
- GPIO and digital I/O
- Timers and counters
- Interrupts and ISRs

TEACHING FOCUS

Program the processor and respond to events.

KEY TAKEAWAY Interrupts make embedded systems responsive.

EMBEDDED SYSTEMS

3

SESSION

60 MINUTES

ANALOG & COMMUNICATION

KEY TOPICS

- ADC and DAC
- PWM for motor control
- UART, I2C, SPI protocols
- Sensor interfacing

TEACHING FOCUS

Bridge the digital processor and analog world.

KEY TAKEAWAY Protocols let chips talk to sensors and actuators.

EMBEDDED SYSTEMS

4

SESSION

 60 MINUTES

REAL-TIME CONCEPTS

KEY TOPICS

- Determinism and latency
- Task scheduling basics
- RTOS introduction
- Deadlines and jitter

TEACHING FOCUS

Guarantee timing-critical robot behavior.

▲ **KEY TAKEAWAY** In robotics, late answers can be wrong answers.

EMBEDDED SYSTEMS

5

SESSION

 60 MINUTES

POWER & HARDWARE INTEGRATION

KEY TOPICS

- Power budgets and regulators
- PCB layout basics
- Debugging embedded systems
- Hardware–software co-design

TEACHING FOCUS

Build reliable embedded hardware.

▲ **KEY TAKEAWAY** Power and layout determine reliability.

EMBEDDED SYSTEMS

6

SESSION

60 MINUTES

EMBEDDED AI & REVIEW

KEY TOPICS

- TinyML and on-device inference
- Course recap and assessment
- Toward IoT (AI.201)
- Final project preview

TEACHING FOCUS

Run intelligence on the edge.

KEY TAKEAWAY AI on the edge means intelligence wherever the robot